Coherent, regular, simple, passive: a reconciliation of the two decomposition traditions

(Rosenfeld–Gröbner vs. differential Thomas)

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Abstract

A companion note to *A New Method of Solving PDEs*, recording the exact relationship between four notions that recur whenever one compares the Rosenfeld–Gröbner (RG) and differential-Thomas decompositions: a differential triangular system may be *coherent*, *regular*, *(differentially) simple*, or *passive* (= *Janet-involutive*). These are *not* synonyms and they are *not* mutually equivalent. They form a strict one-directional hierarchy

$$\text{simple} \implies \text{regular} \implies \text{coherent}, \qquad\qquad \text{passive} \implies \text{coherent},$$

with coherence as the common base property ("all Δ-polynomials reduce to zero") that makes Rosenfeld's Lemma fire. The note pins each node to a primary source, quotes the defining conditions verbatim, and flags one correction to an earlier informal claim (that Seiler's monograph states a coherence $\Leftrightarrow$ involution *equivalence*): the sourced statement is the *implication* passive $\Rightarrow$ coherent (Li–Wang 1999, Thm. 3; Gerdt 1999, Thm. 16; Bächler et al. 2012).

1 Why this note exists

RG and differential Thomas both decompose the radical differential ideal $\{\mathbb{P}\}$ generated by a finite set $\mathbb{P}$ of differential polynomials, but they return components of different *kinds*:

- RG returns *regular* differential systems and represents $\{\mathbb{P}\}$ as an *intersection* of regular differential ideals (components may overlap);
- differential Thomas returns *simple* differential systems and gives a *disjoint* decomposition (the solution sets partition).

A duality caveat. The two bullets above are stated at *different* levels, and it is worth being explicit about it, because "intersection" and "union" are dual here. RG's "intersection" is at the *ideal* level, $\sqrt{[\mathbb{P}]} = \bigcap_k \sqrt{[A_k] : H_k^\infty}$; dually, at the *variety* level it is a *union* d-Zero$(\mathbb{P})$ = $\bigcup_k$ d-Zero(regular$_k$) of the component solution sets. Thomas's "disjoint decomposition" is stated at the *variety* level. So the honest common frame is: *both* are unions of solution sets; RG's is a possibly-*overlapping* union (equivalently, an ideal intersection), while Thomas's is a *disjoint* union (a partition). The "intersection vs. disjoint-union" contrast is really "ideal-intersection (RG) vs. variety-partition (Thomas)", which compares across the duality; each statement is individually the conventional one for its algorithm.

Where parametric RG sits: a union of intersections. A third variant, *parametric* RG (Fakouri–Rahmany–Basiri 2018; the algorithm ported in `~/parametric_rg`), adds a *disjoint* stratification of parameter space on top of an RG decomposition of the differential fibre. Writing the parameter space as a partition into cells $\Omega_{\text{par}} = \bigsqcup_j C_j$,

$$\text{d-Zero}(\mathbb{P}) \;=\; \bigsqcup_j \Big(\underbrace{\textstyle\bigcup_k \text{d-Zero}(\text{regular}_{j,k})}_{\text{RG decomposition over cell } C_j} \Big),$$

or, naming each cell's fibre by its ideal representation, over C_j it is $\bigcap_k \sqrt{[A_{j,k}] : H^\infty}$. In words:

$$\boxed{\text{parametric RG} \;=\; \text{a } \textit{disjoint} \text{ union (over parameter cells) of RG intersections (per cell)}}$$

— the outer union disjoint (Thomas-like, *on the parameter axis*), the inner object an ordinary RG intersection of regular components (RG-like, *on the differential fibre*, and possibly overlapping as varieties). It is a union *of* intersections, never an intersection of unions: across disjoint parameter cells one can only case-split, never intersect. This is the precise sense of the "hybrid" character noted in §6 — disjoint on parameters, overlapping-regular on the fibre.

Underneath all of these sits a third notion, *coherence* (the condition in Rosenfeld's Lemma), and on the Janet–Riquier side a fourth, *passivity / involution*. In loose usage these four words get treated as interchangeable. They are not. The purpose of this note is to fix the definitions from primary sources and record the exact implication lattice, so the distinction need not be rederived. The single most on-point reference is Li & Wang (1999), whose very title is the question — "*Coherent, regular and simple systems in zero decompositions of partial differential systems*" — and whose abstract states its purpose as: *"Rosenfeld's lemma is extended to a more general setting; relations between passivity and coherence are clarified; regular systems and simple systems are generalized and proposed."*

2 Standing notation

Fix a differential field K of characteristic 0 with commuting derivations, and the differential polynomial ring $\mathfrak{R} = K\{X_1, \ldots, X_n\}$. For a differential polynomial $P \in \mathfrak{R} \setminus K$: $\mathrm{ld}(P)$ is its *lead* (highest-ranked derivative under a fixed admissible ranking), $\mathrm{ini}(P)$ its *initial* (leading coefficient in $\mathrm{ld}(P)$), and $\mathrm{sep}(P) = \partial P / \partial\, \mathrm{ld}(P)$ its *separant*. Write $\mathrm{si}(\mathbb{P})$ for the set of separants and initials of $\mathbb{P}$, and $[\mathbb{P}]$, $\{\mathbb{P}\}$ for the differential and radical differential ideals generated by $\mathbb{P}$. Saturation: $[\mathbb{P}] : \mathrm{si}(\mathbb{P})^\infty$ denotes the differential ideal quotient by the multiplicative set generated by $\mathrm{si}(\mathbb{P})$. $\text{d-Zero}(\mathbb{P}/\mathbb{Q})$ is the differential zero set of $\mathbb{P}$ off the zero set of $\mathbb{Q}$; d-prem is the differential pseudo-remainder.

We use "d-tri set" for a differentially triangular set (distinct leads, pairwise partially reduced) and "d-asc set" for a differential ascending (autoreduced) set. Every d-asc set is a d-tri set; the converse fails, and Li & Wang work with the weaker d-tri notion throughout.

3 The four definitions, from the sources

3.1 Coherence (Li–Wang 1999, Def. 5)

The Δ-polynomial of $F, G \in \mathfrak{R} \setminus K$ of the same class is the differential analogue of an S-polynomial (construction due to Rosenfeld):

$$\Delta(F, G) = \mathrm{sep}(G)\, \phi_1 F - \mathrm{sep}(F)\, \phi_2 G, \qquad \mathrm{ld}(\phi_1 F) = \mathrm{ld}(\phi_2 G), \tag{1}$$

where ϕ_1, ϕ_2 are proper derivative operators of *lowest* order bringing the two leads to a common derivative.

> **Definition (coherent).** A weakly-reduced set $\mathbb{P}$ is *coherent* if, for any $F, G \in \mathbb{P}$ of the same class and any $\Delta(F, G)$ as in (1),
>
> $$J\,\Delta(F,G) = C_1\theta_1 P_1 + \cdots + C_r\theta_r P_r, \qquad \theta_i P_i < \mathrm{ld}(\phi_1 F),$$
>
> for some SI-product J, coefficients $C_i \in \mathfrak{R}$, and $P_i \in \mathbb{P}$.

In words: *every* Δ*-polynomial reduces to a combination of lower-lead derivatives of members of* $\mathbb{P}$. This is precisely the differential Buchberger criterion. ("Weakly reduced" (Def. 3) means every $\mathrm{ini}(P)$ and $\mathrm{sep}(P)$ is partially reduced w.r.t. $\mathbb{P}$; every d-tri set is weakly reduced.)

What coherence buys: the generalized Rosenfeld Lemma. *Li–Wang Theorem 2.* If $\mathbb{P} \subset \mathfrak{R} \setminus K$ is weakly reduced and coherent, and $F \in [\mathbb{P}] : \mathrm{si}(\mathbb{P})^\infty$ is partially reduced w.r.t. $\mathbb{P}$, then $F \in (\mathbb{P}) : \mathrm{si}(\mathbb{P})^\infty$. That is: coherence is exactly the hypothesis under which *differential* ideal membership collapses to *algebraic* ideal membership. This is the theoretical engine of the entire RG algorithm (Boulier–Lazard–Ollivier–Petitot).

3.2 Regular (Li–Wang 1999, Def. 6)

First, the system-level upgrade of coherence: a d-tri system $\langle \mathbb{T}, \mathbb{U} \rangle$ is *coherent* if $\mathbb{T}$ is coherent and each d-pol in $\mathbb{U}$ is partially reduced w.r.t. $\mathbb{T}$. Then:

> **Definition (regular).** A d-tri system $\langle \mathbb{T}, \mathbb{U} \rangle$ is *regular* if it is a coherent *and* elementarily-triangular (e-tri) system.

$$\boxed{\text{regular} \ = \ \text{coherent} \ + \ \text{e-tri}}$$

Here $\langle \mathbb{T}, \mathbb{U} \rangle$ is *e-tri* (Def. 1) if *every* point in $\mathrm{Zero}(\mathbb{T}/\mathbb{U})$ — the *algebraic*, not just differential, zero set — fails to annihilate any element of $\mathrm{si}(\mathbb{T})$; equivalently the initials and separants are invertible on the whole (algebraic) component, not merely generically. Li & Wang note this *generalizes* Boulier et al.'s regular chains in two ways: $\mathbb{T}$ need not be a d-asc set, and $\mathrm{si}(\mathbb{T})$ need not be contained in $\mathbb{U}$.

What regularity buys. *Li–Wang Theorem 8.* For a regular $\langle \mathbb{T}, \mathbb{U} \rangle$: a d-pol F lies in $[\mathbb{T}] : \mathbb{U}^\infty$ iff a partial pseudo-remainder of F w.r.t. $\mathbb{T}$ lies in $(\mathbb{T}) : \mathbb{U}^\infty$; and both $[\mathbb{T}] : \mathbb{U}^\infty$ and $(\mathbb{T}) : \mathbb{U}^\infty$ are *radical.* Regular systems are the components of the RG "weak form" $\text{d-Zero}(\mathbb{P}/\mathbb{Q}) = \bigcup_i \text{d-Zero}(\mathbb{T}_i/\mathbb{U}_i)$ — an *intersection*-style decomposition whose components can overlap.

3.3 Differentially simple (Li–Wang 1999, Def. 8; algebraic core Def. 2)

The algebraic prototype (Def. 2, after Thomas and Wang): a triangular system $\langle \mathbb{T}, \mathbb{U} \rangle$ in $K[\mathbf{x}]$ is *w-simple* if for each $1 \le k \le n$ it is triangular, has non-vanishing initials on the lower solution set, and is *square-free* (the univariate image $T(\bar{\mathbf{x}}^{\{p-1\}}, x_p)$ is square-free). Lifting to the differential setting:

> **Definition (differentially simple, d-sim).** A *coherent* d-tri system $\langle \mathbb{T}, \mathbb{U} \rangle$ in $\mathfrak{R}$ is *differentially simple* if, when considered as a triangular system in the underlying (non-differential) ring, it is w-simple.

So, informally,

$$\text{simple} \;=\; \text{regular} \;+\; \text{square-free / w-simple conditions.}$$

The disjoint decomposition $\text{d-Zero}(\mathbb{P}/\mathbb{Q}) = \bigcup_i \text{d-Zero}(\mathbb{T}_i/\mathbb{U}_i)$ into d-sim systems is the (differential) *Thomas decomposition*; its components *partition* the solution set.

What simplicity buys. *Li–Wang Theorem 9 & Corollary 2.* A d-sim system has $\text{d-Zero}(\mathbb{T}/\mathbb{U}) \neq \emptyset$, and for any d-pol P,

$$\text{d-Zero}(\mathbb{T}/\mathbb{U}) \subset \text{d-Zero}(P) \iff \text{d-prem}(P, \mathbb{T}) = 0.$$

Moreover $\mathbb{T}$ is then a characteristic set (in Ritt's sense) of the prime differential ideal $[\mathbb{T}] : \mathbb{U}^\infty$. And, top of Li–Wang p. 58: "*a d-sim system must be regular.*"

3.4 Passive / involutive (Gerdt 1999; used by differential Thomas)

"Passive" is the Janet–Riquier word for the same completeness idea, phrased via an *involutive division* (Janet division for the Thomas construction). Gerdt states the terminology bridge in prose: *"A system satisfying the Janet involutivity conditions is often called **passive**."* The condition itself:

> **Definition (involutive, Gerdt 1999 Def. 14).** An L-autoreduced set F is L-*involutive* if
>
> $$(\forall f \in F)\,(\forall \alpha \in \mathbb{N}^n)\;\big[\,\mathrm{NF}_L(\partial_\alpha f,\, F) = 0\,\big].$$
>
> **Local criterion (Gerdt 1999 Thm. 16).** F is L-involutive (for a continuous involutive division L) iff
>
> $$(\forall f \in F)\,(\forall x_i \in \mathrm{NM}_L(f, F))\;\big[\,\mathrm{NF}_L(\partial_{x_i} \cdot f,\, F) = 0\,\big],$$
>
> i.e. only the *non-multiplicative* prolongations need reduce to zero.

This finite, checkable criterion is "the involutivity criterion" cited (in its nonlinear-PDE form, Gerdt 2008) by Bächler et al. (2012). Note the exact match with the involutive clause in the Bächler et al. definition of a differential simple system: *"S is (Janet) involutive if all non-admissible prolongations in $(S_T)^=$ reduce to zero by $(S_T)^=$."* Gerdt also records *Theorem 17*: an L-involutive basis of $[F]$ is a *differential Gröbner basis*; and *Theorem 18*, an involutive analogue of the Buchberger chain criterion — the same S-polynomial / Δ-polynomial content that, on the RG side, *is* coherence.

4 The connecting theorem: passive ⇒ coherent

The bridge between the two vocabularies is a single, one-directional theorem, stated and proved in Li–Wang and echoed in the Thomas literature.

> **Li–Wang 1999, Theorem 3.** *A passive d-asc set is coherent.*

The proof (Li–Wang §4.1, "Passivity and coherence") runs through Wu's tuple-decomposition and passivity theorems; the section's framing is explicit that this direction is the useful one: "*coherent d-asc sets are, in some sense, simpler than passive d-asc sets.*" The converse is *not* claimed.
The same implication is stated operationally in the differential-Thomas paper: Bächler, Gerdt, Lange-Hegermann & Robertz (2012), §1: "*the output systems are Janet-involutive … and **hence they are coherent***," and, on the algorithm, "*the involutive approach, which we utilize to make the subsystems coherent.*" Direction: involutive ⇒ coherent, via Gerdt's involutivity criterion.

Correction to an earlier informal claim. It is tempting (and was asserted in passing during the discussion that produced this note) to attribute a coherence $\Leftrightarrow$ involution *equivalence* to W. Seiler's monograph *Involution* (2010) and, worse, to his two open-access AAECC 2009 papers ("A combinatorial approach to involution and δ-regularity, I & II"). This is not supported by those two papers: they are commutative-algebra papers on involutive/Pommaret bases and δ-regularity and contain *zero* occurrences of "coherence," "Rosenfeld," or "regular differential." They connect involution to Janet–Riquier *passivity*, not to RG coherence. The defensible, sourced statement is the *implication* above, not an equivalence, and its primary home is Li–Wang 1999 (Thm. 3) and Gerdt 1999/2008, not Seiler. (Seiler's δ-regularity work is nonetheless the right reference for *why* Pommaret-involution — unlike Janet-involution and unlike coherence — is coordinate-dependent, which is a second, independent reason the two notions cannot be identical.)

5 The implication lattice

Collecting Definitions 5, 6, 8 and Theorems 2, 3 of Li–Wang with Gerdt's Theorem 16:

$$\underbrace{\text{passive}}_{\substack{\text{Janet-involutive}\\ \text{Gerdt Def. 14/Thm. 16}}} \xRightarrow{\text{L–W Thm. 3}} \text{coherent}$$

$$\underbrace{\text{simple}}_{\text{L–W Def. 8}} \Longrightarrow \underbrace{\text{regular}}_{\text{L–W Def. 6}} = \text{coherent} + \text{e-tri} \Longrightarrow \text{coherent.}$$

Node	Defining condition	Decomposition	Source
coherent	Δ-polys reduce to 0	(property)	L–W Def. 5
passive / involutive	non-mult. prolongations reduce to 0	(property)	Gerdt Def. 14, Thm. 16
regular	coherent + e-tri (sep./init. invertible on the component)	intersection (RG weak form)	L–W Def. 6
simple (d-sim)	regular + w-simple (triangular, non-vanishing init., square-free)	disjoint (Thomas)	L–W Def. 8

Every arrow is *strict* and *one-directional*:

- **coherent** is the base property — it makes Rosenfeld's Lemma fire (differential $\to$ algebraic), and nothing more is assumed.
- **passive** $\Rightarrow$ **coherent** but not conversely: passivity is a specific, division-dependent (Janet) completion; coherence is the weaker, division-free, intrinsic property it implies (L–W Thm. 3).
- **regular** $\Rightarrow$ **coherent** but not conversely: regular adds the e-tri (whole-component invertibility of separants/initials) upgrade, which is what makes the saturated ideals radical (L–W Thm. 8).
- **simple** $\Rightarrow$ **regular** but not conversely: simple adds square-freeness / the w-simple conditions, which is what buys *disjointness* of the Thomas decomposition and the characteristic-set property (L–W Cor. 2).

6 Consequences for the RG-vs-Thomas comparison

- **Coherence is not the discriminator between RG and Thomas.** Both traditions require it (RG via Rosenfeld's Lemma; Thomas via the involutive clause that *implies* it). The real discriminators are the *extra* conditions: e-tri (regular) vs. e-tri + square-free (simple), i.e. *overlapping intersection* vs. *disjoint partition.*

- **Δ-polynomial = differential S-polynomial.** Bächler et al. (2012) say it outright ("the analogon of s-polynomials in differential algebra"), and Gerdt's Theorem 18 is the involutive Buchberger chain criterion. So the RG/Thomas relationship mirrors the algebraic Gröbner-basis / involutive-basis relationship: every involutive basis is a Gröbner basis but generally larger and more structured; likewise passive $\Rightarrow$ coherent, and simple is the more refined (disjoint) decomposition.

- **On "is a differential Thomas decomposition always finer than a (parametric) RG decomposition?"** Fibrewise (same ranking, differential structure), yes: simple $\Rightarrow$ regular, and simple exposes the degenerate branches RG saturation folds away. But *globally*, including a parameter stratification, the two can become *incomparable*, because they split along the discriminant loci that *their own* orderings produce, and parametric RG carries a distinguished parameter layer (with deferred genericity conditions) that Thomas — having "no notion of parameter" — does not reproduce. The one-directional lattice here is what holds *after* fixing a common ambient ring and ranking; it is not a claim of global refinement.

7 Primary sources

- Z. Li & D. Wang, *Coherent, regular and simple systems in zero decompositions of partial differential systems*, Systems Science and Mathematical Sciences **12** (Suppl.), May 1999, 43–60. [Defs. 5 (coherent), 6 (regular), 8 (d-sim); Thm. 2 (generalized Rosenfeld), Thm. 3 (passive $\Rightarrow$ coherent), Thm. 8 (radicality), Thm. 9/Cor. 2 (simple).] Local archive: `~/project/papers/li-wang-1999-coherent-regular-simple-systems-zero-decompositions-pde.pdf` (scanned; page images read directly).

- V. P. Gerdt, *Completion of Linear Differential Systems to Involution*, CASC 1999, 115–137 (arXiv:math/9909114). [Def. 14 (involutive), Thm. 16 (local involutivity criterion: non-multiplicative prolongations reduce to 0), Thm. 17 (involutive $\Rightarrow$ differential Gröbner), Thm. 18 (involutive Buchberger chain criterion); prose: passive = Janet-involutive.] Local archive: `~/project/papers/gerdt-1999-completion-linear-differential-systems-to-involution.pdf`.

- V. P. Gerdt, *On Decomposition of Algebraic PDE Systems into Simple Subsystems*, Acta Appl. Math. **101** (2008), 39–51. [The involutivity criterion in nonlinear-PDE form, as cited by Bächler et al.]

- T. Bächler, V. Gerdt, M. Lange-Hegermann & D. Robertz, *Thomas Decomposition of Algebraic and Differential Systems*, J. Symbolic Comput. **47** (2012), 1233–1266 (arXiv:1008.3767). [Def. 2.1 (algebraic simple), Def. 3.3 (differential simple = algebraically simple + involutive + minimal + inequation-reduced); "Janet-involutive ... hence coherent"; Δ-poly = s-poly analogue.] Local archives (published + preprint) in `~/project/papers/`.

- W. M. Seiler, *A Combinatorial Approach to Involution and δ-Regularity, I & II*, AAECC **20** (2009). [Involutive / Pommaret bases, δ-regularity. *Not* a source for a coherence $\Leftrightarrow$ involution equivalence — see §4 correction. Relevant for the coordinate-dependence of Pommaret involution.]

Note prepared as computational/literature backing for NewMethod; reconciles the terminology used across the RG and differential-Thomas ports (`joca.sage`, `joca-thomas-native.sage`, and the parametric-RG port).